HTTP Methods

GET Read only

تستخدم لجلب البيانات فقط من السيرفر.

GET /users

Data in the URL

characteristics

1. Idempotent ✓
2. Cacheable ✓
3. Safe ✓

EX

```
GET
/login?username=admin&password=123456
HTTP/1.1   Host: example.com
```

This Part is A Query Parameter

username=admin&password=123456

Sensitive Data Exposure

POST Create / send data

تستخدم لإرسال بيانات جديدة للسيرفر.

POST /users

```
{
"name": "Mahmoud",
"age": 22
}
```

characteristics

1. Idempotent ✗
2. Cacheable ✗
3. Safe ✗

Injection attacks — without Validation

```
POST /login
{
"username": "' OR 1=1--",
"password": "x"
}
```

PUT Update

تستخدم لتحديث البيانات بالكامل.

PUT /users/5

```
{
"name": "Mahmoud",
"age": 23
}
```

characteristics

1. Idempotent ✓
2. Safe ✗

PATCH Update

تستخدم لتحديث جزء من البيانات فقط.

PATCH /users/5

```
{
"email": "new@mail.com"
}
```

DELETE Remove

تستخدم لحذف البيانات.

DELETE /users/5

characteristics

1. Idempotent ✓
2. Safe ✗

OPTIONS Method + CORS Flow

Server tells the client which methods are allowed

GET
POST
PUT
DELETE

request:

OPTIONS /api/users HTTP/1.1
Host: example.com
Origin: https://app.com
Access-Control-Request-Method: PUT

Pre-flight request

Response:

HTTP/1.1 204 No Content
Access-Control-Allow-Origin: https://app.com
Access-Control-Allow-Methods: GET, POST, PUT, DELETE

Client

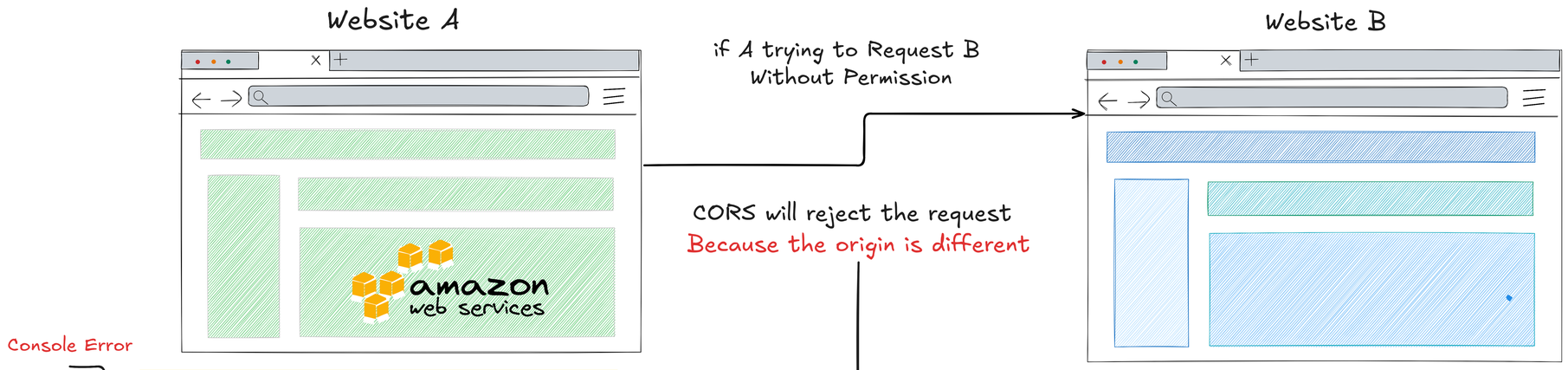
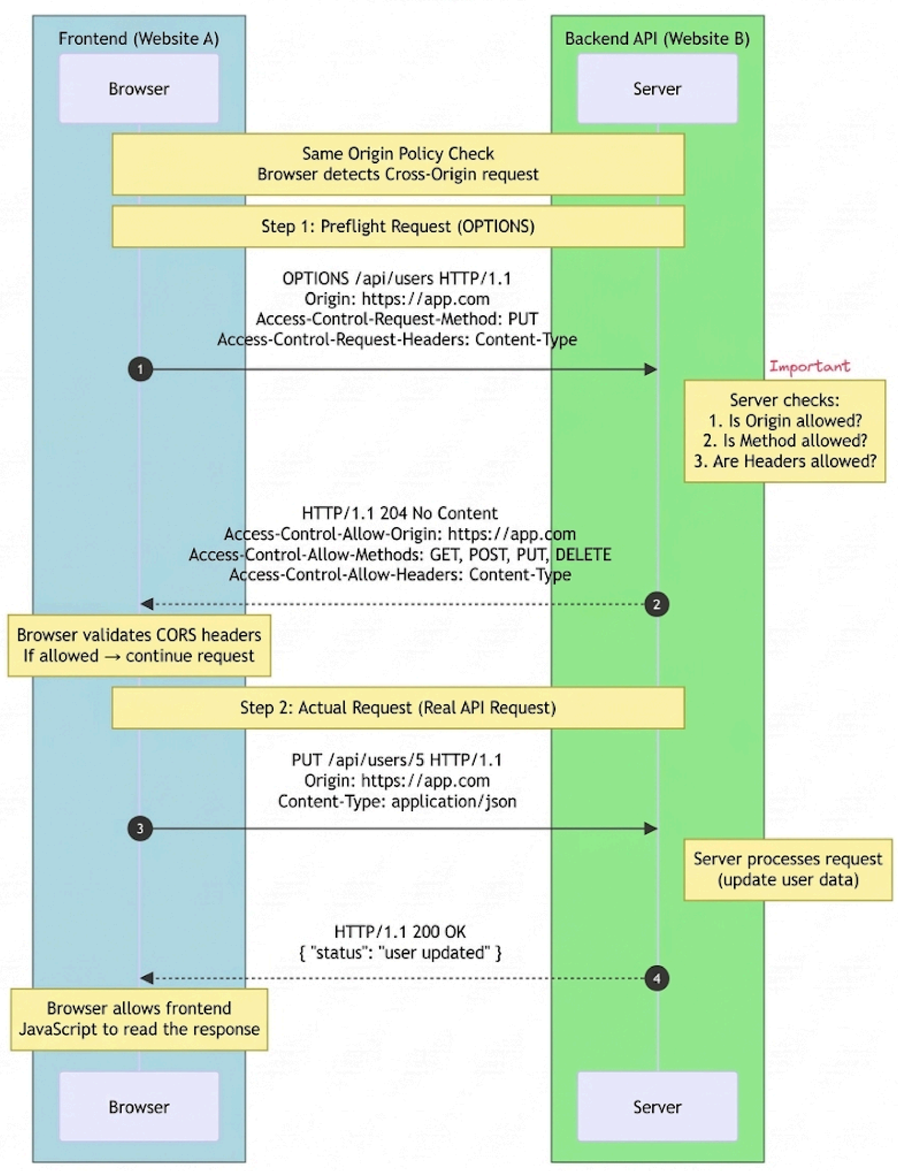
Server

Cross-Origin Resource Sharing

CORS

security feature implemented by browsers to control how resources on a web page can be requested from another domain (origin) outside the domain from which the first resource was served

Origin = Protocol + Domain + Port



CORS Preflight Flow